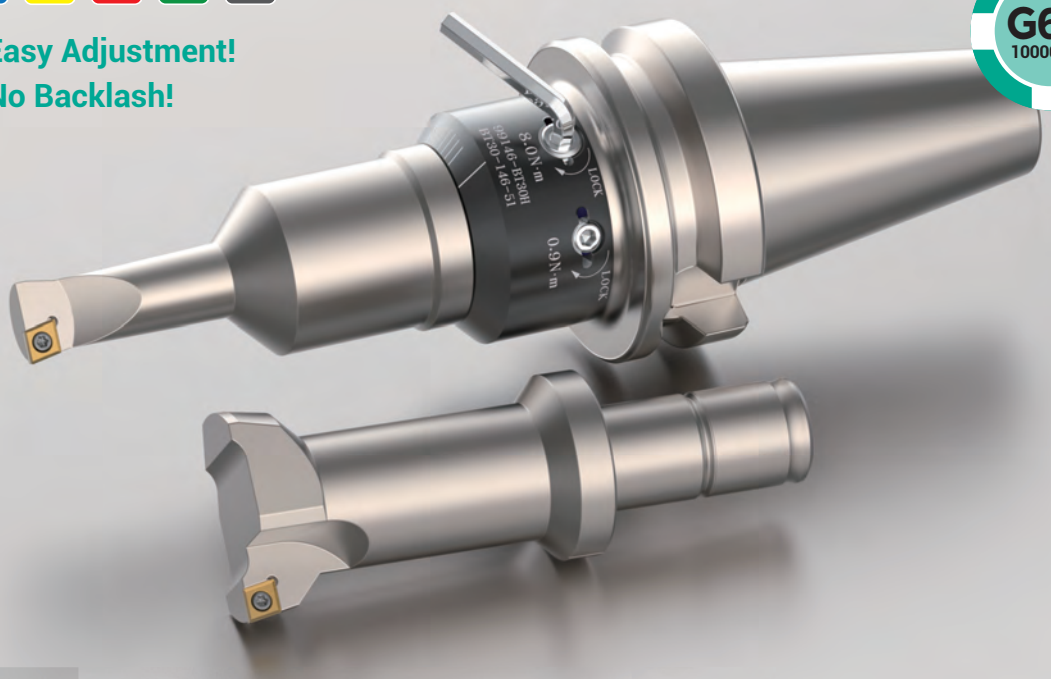
NineBore >>>

Cycle Time
Roughness
Position Accuracy
True Roundness

Nine9 offers a range of eccentrically mechanism boring tools which designed to provide stability, accuracy, and productivity in boring operations.
These tools are commonly used in industries such as automotive, aerospace, mold and die, and general machining.
99146 can complete rough and finish boring in one operation.

P M K N H

G6.3
10000rpm





- Easy Adjustment!
- No Backlash!

Features >>>

Easy Handling

- Change the boring bar in just one minute.
- Dimensions are easy to read. They are indicated on the tools and are easily adjusted on a tool presetter or in machining center.
- Adjusting range :±0.1mm

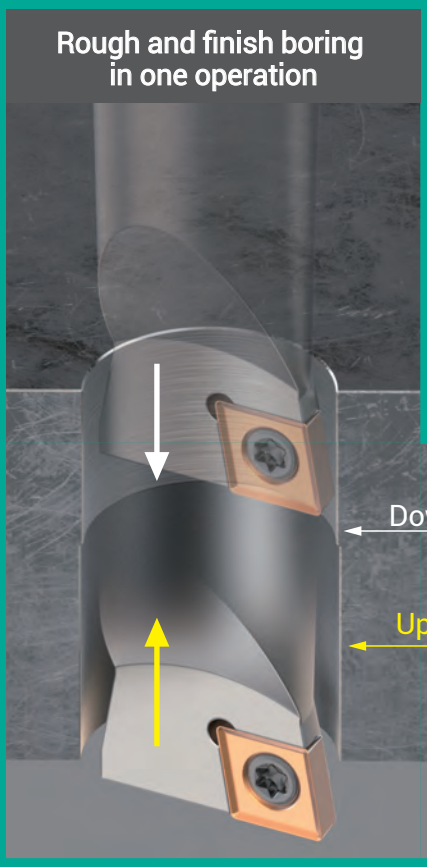
High Speed.

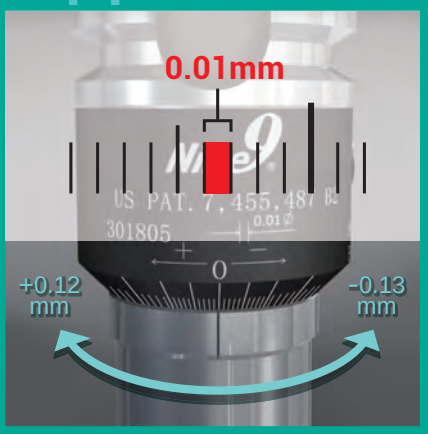
- Good for fine boring operation on milling machines, machining centers and special purpose machines.
- Replace solid carbide reamers.

99146 Series Is Ideal For Casting Aluminum With Uncertain Pre-Hole Dimensions And Deviations In Hole Spacing.

Applications

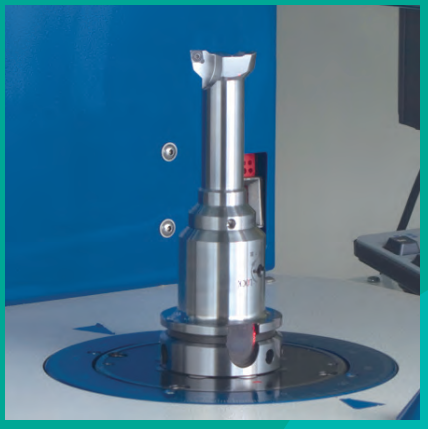






“Ø5~Ø50mm boring bars are interchangeable, balanced grade is G6.3 10000 r.p.m., all sizes are guaranteed.”

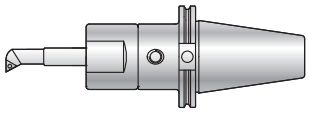




NineBore Features

Precisely pre-balanced the basic holder and the boring bar >>

• Balanced grade is maintained while adjusting boring dimension or change to different diameter boring bars.

NineBore design	Other Boring Tool
	
• Pre-balanced	• Without pre-balanced

Application >>

• Ideal as small hole boring tool with excellent accuracy.
• For fine boring operation on milling machines, machining centres and special purpose machines.

High Speed >>

• Boring bar design ensures accurate high speed boring.
Balanced grade is G6.3 10000 r.p.m., all sizes are guaranteed.
• Combination bore / chamfer / facing / multi-task tools can be ordered on request.

Economic >>

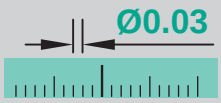
• Low cost micro adjustable boring heads.
• It can replace end mill and brazed tool bits.
• This simple boring tool has minimal components. In minutes, the boring bar can be changed and the boring dimension set on the tool presetter.

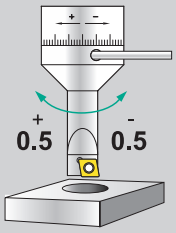
Work Task HoleØ : 10mm (H7 Tolerance) Hole Distance : 20mm		
Tool	NineBore 99146	Other Boring Tool
Workpiece material	N AL6061T6	
CNC Code	G85	G76
Spindle speed r.p.m.	**10000 r.p.m.**	2500 r.p.m.
Feed rate f = mm/rev.	0.07 mm/rev.	0.07 mm/rev.
Feed rate F= mm/min.	**700 mm/min.**	175mm/min.
	RESULT	
Drilling time sec.	**4s 75**	10s 44
Hole 1 Ø mm	Ø10.006 mm	Ø10.003 mm
Hole 2 Ø mm	Ø10.005 mm	Ø9.990 mm
Hole Distance mm	Ø20.00 mm	Ø19.98 mm
Comparison		

NineBore System

Series	Adjustment range	ØD range	Insert	Boring Bars	Head Shank
99101 Each Division 0.03 mm/div	±0.5mm	Ø 6.5 mm ≀ 25.5 mm	CCGT030102 CCGT040102 CCGH060204	00-99101-07...25	Side lock holders Hydraulic chucks Spring collet chucks
99121 Each Division 0.01 mm/div.	±0.1mm	Ø 4.9 mm ≀ 25.1 mm	CCGT030102 CCGT040102 CCGH060204	00-99121-05...25	
G6.3 10000 r.p.m **99146** Each Division 0.01 mm/div.	+0.12mm ≀ -0.13mm	Ø 4.87 mm ≀ 50.12 mm	CCGT030102 CCGT040102 CCGH060204	ØD±0.1 Ø5 ~ Ø10 mm 00-99146-xxS ØD±0.1 Ø11 ~ Ø25 mm 00-99146-xxS ØD±0.1 Ø26 ~ Ø39 mm 00-99146-xxA ØD±0.1 Ø40 ~ Ø50 mm 00-99146-xxA	00-99146-BT30H 00-99146-BT40H 00-99146-BT50H 00-99146-CAT40H
4~6XD **99151** Each Division 0.01 mm/div.	+0.12mm ≀ -0.13mm	Ø 4.87 mm ≀ 20.12 mm	CCGT030102 CCGT040102 CCGH060204	Ø4.87 ~ Ø6.87 mm Ø6 Ø6.87 ~ Ø8.87 mm Ø8 Ø8.87 ~ Ø11.12 mm Ø10 Ø11.87 ~ Ø14.12 mm Ø11 Ø14.87 ~ Ø20.12 mm Ø15.5 Adapter Ø 00-99151A-06...15.5	00-99146-HSK63AH 00-99146-SB32H

99101 EMB Boring Bars 0.03 mm/div.

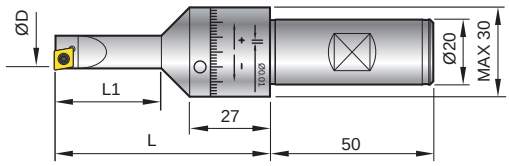




Boring Bars >>

- Adjusted to required diameter by eccentric mechanism
- H type with internal coolant can be ordered on request from Dia. 7mm. Ordering example: 00-99101-07H.
- Other sizes are available on request.

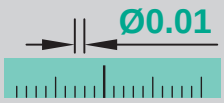


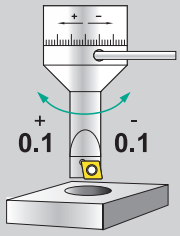


Part No.	Type		ØD	L1	L	Insert	Screw / Key
00-99101-07	SB20-0721-03		6.5-7.5	21	60	CC...040102	*NS-20036 0.6Nm / NK-T6
00-99101-08	SB20-0824-03		7.5-8.5	24	63		
00-99101-09	SB20-0927-03		8.5-9.5	27	65	CC...0602...	*NS-25045 0.9Nm / NK-T7
00-99101-10	SB20-1030-03		9.5-10.5	30	68		
00-99101-11	SB20-1133-03		10.5-11.5	33	70		
00-99101-12	SB20-1236-03		11.5-12.5	36	73		
00-99101-13	SB20-1339-03		12.5-13.5	39	75		
00-99101-14	SB20-1442-03		13.5-14.5	42	78		
00-99101-15	SB20-1545-03	Adjustment range:±0.5mm Each Division 0.03mm	14.5-15.5	45	80		
00-99101-16	SB20-1648-03		15.5-16.5	48	83	CC...0602...	*NS-25060 0.9Nm / NK-T7
00-99101-17	SB20-1751-03		16.5-17.5	51	85		
00-99101-18	SB20-1850-03		17.5-18.5	50	82		
00-99101-19	SB20-1950-03		18.5-19.5	50	82		
00-99101-20	SB20-2050-03		19.5-20.5	50	82		
00-99101-21	SB20-2150-03		20.5-21.5	50	82		
00-99101-22	SB20-2250-03		21.5-22.5	50	82		
00-99101-23	SB20-2350-03		22.5-23.5	50	82		
00-99101-24	SB20-2450-03		23.5-24.5	50	82		
00-99101-25	SB20-2550-03		24.5-25.5	50	82		

*Torque screwdriver is recommended.

99121 EMB Boring Bars 0.01 mm/div.

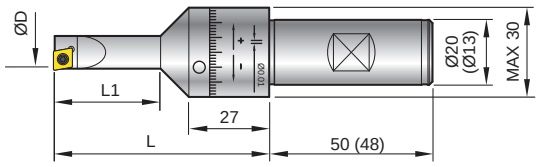




Boring Bars >>

- Adjusted to required diameter by eccentric mechanism
- H type with internal coolant can be ordered on request from Dia. 7mm. Ordering example: 00-99121-07H.
- Other sizes are available on request.

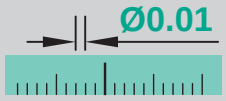




Part No.	Type		ØD	L1	L	Insert	Screw / Key
00-99121-05	SB20-0515-01		4.9-5.1	15	54	CC...030102	*NS-16030 0.4Nm / NK-T6
00-99121-06	SB20-0618-01		5.9-6.1	18	57		
00-99121-07	SB20-0721-01		6.9-7.1	21	60	CC...040102	*NS-20036 0.6Nm / NK-T6
00-99121-08	SB20-0824-01		7.9-8.1	24	63		
00-99121-09	SB20-0927-01		8.9-9.1	27	65	CC...0602...	*NS-25045 0.9Nm / NK-T7
00-99121-10	SB20-1030-01		9.9-10.1	30	68		
00-99121-11	SB20-1133-01		10.9-11.1	33	70		
00-99121-12	SB20-1236-01		11.9-12.1	36	73		
00-99121-13	SB20-1339-01		12.9-13.1	39	75		
00-99121-14	SB20-1442-01	Adjustment range:±0.1mm	13.9-14.1	42	78		
00-99121-15	SB20-1545-01		14.9-15.1	45	80		
00-99121-16	SB20-1648-01	Each Division 0.01mm	15.9-16.1	48	83	CC...0602...	*NS-25060 0.9Nm / NK-T7
00-99121-17	SB20-1751-01		16.9-17.1	51	85		
00-99121-18	SB20-1850-01		17.9-18.1	50	82		
00-99121-19	SB20-1950-01		18.9-19.1	50	82		
00-99121-20	SB20-2050-01		19.9-20.1	50	82		
00-99121-21	SB20-2150-01		20.9-21.1	50	82		
00-99121-22	SB20-2250-01		21.9-22.1	50	82		
00-99121-23	SB20-2350-01		22.9-23.1	50	82		
00-99121-24	SB20-2450-01		23.9-24.1	50	82		
00-99121-25	SB20-2550-01		24.9-25.1	50	82		

*Torque screwdriver is recommended.

99146 Quick Change High Speed EMB Boring Bar



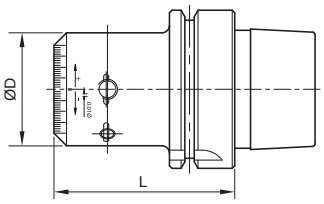




Boring Head Shank >>

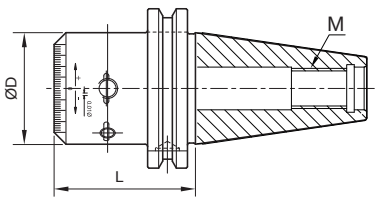
- HSK63





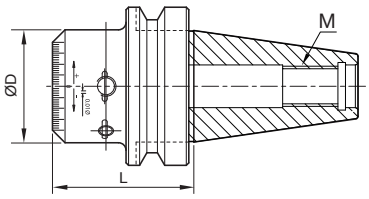
- CAT40





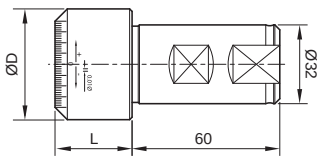
- BT



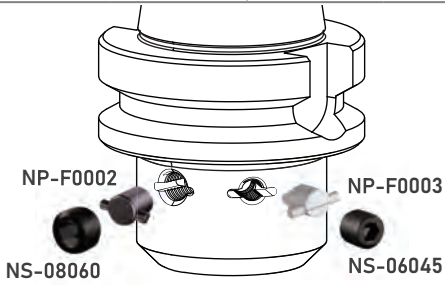


- SB32



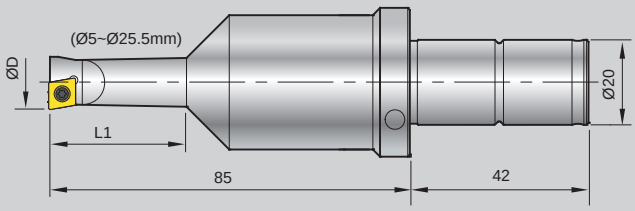


Part No.	Type	ØD	L	M	M8 Screw		M6 Screw	
					Spring Fingers	Locking Screw	Spring Fingers	Pre-load Screw
00-99146-HSK63AH	HSK63A-146-72	45	72.0	-	NP-F0002	NS-08060 8.0 Nm	NP-F0003	NS-06045 0.9 Nm
00-99146-CAT40H	CAT40-146-56	45	56.3	5/8"-11				
00-99146-BT30H	BT30-146-51	45	51.3	M12xP1.75				
00-99146-BT40H	BT40-146-56	45	56.3	M16xP2.0				
00-99146-BT50H	BT50-146-77	45	77.3	M24xP3.0				
00-99146-SB32H	SB32-146-31	45	31.3	-				



99146 Quick Change High Speed EMB Boring Bar

Boring Bar Ø5~Ø25 >>



- Alloy Steel Shank. Boring Depth : L1, 2~3xD
- H type with internal coolant can be ordered on request from Dia. 10mm.
  Ordering example: 00-99146-1000SH.
- Other sizes are available on request.

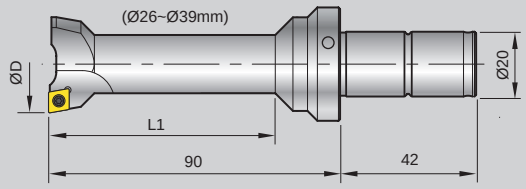
Part No.	Type	ØD	L1	Insert Screw / Key
00-99146-0500S	C20-0500-10L	4.87~5.12	10.00	CC...030102 *NS-16030 0.4Nm / NK-T6
00-99146-0600S	C20-0600-12L	5.87~6.12	12.00	
00-99146-0700S	C20-0700-14L	6.87~7.12	14.00	CC...040102 *NS-20036, 0.6Nm / NK-T6
00-99146-0800S	C20-0800-16L	7.87~8.12	16.00	
00-99146-0900S	C20-0900-18L	8.87~9.12	18.00	CC...0602... *NS-25045 0.9Nm NK-T7
00-99146-1000S	C20-1000-25L	9.87~10.12	25.00	
00-99146-1025S	C20-1025-25L	10.12~10.37		
00-99146-1050S	C20-1050-26L	10.37~10.62	26.25	
00-99146-1075S	C20-1075-26L	10.62~10.87		
00-99146-1100S	C20-1100-27L	10.87~11.12	27.50	
00-99146-1125S	C20-1125-27L	11.12~11.37		
00-99146-1150S	C20-1150-28L	11.37~11.62	28.75	
00-99146-1175S	C20-1175-28L	11.62~11.87		
00-99146-1200S	C20-1200-30L	11.87~12.12	30.00	
00-99146-1225S	C20-1225-30L	12.12~12.37		
00-99146-1250S	C20-1250-31L	12.37~12.62	31.25	
00-99146-1275S	C20-1275-31L	12.62~12.87		
00-99146-1300S	C20-1300-32L	12.87~13.12	32.50	
00-99146-1325S	C20-1325-32L	13.12~13.37		
00-99146-1350S	C20-1350-33L	13.37~13.62	33.75	
00-99146-1375S	C20-1375-33L	13.62~13.87		
00-99146-1400S	C20-1400-35L	13.87~14.12	35.00	
00-99146-1425S	C20-1425-35L	14.12~14.37		
00-99146-1450S	C20-1450-36L	14.37~14.62	36.25	
00-99146-1475S	C20-1475-36L	14.62~14.87		
00-99146-1500S	C20-1500-37L	14.87~15.12	37.50	
00-99146-1525S	C20-1525-37L	15.12~15.37		
00-99146-1550S	C20-1550-38L	15.37~15.62	38.75	
00-99146-1575S	C20-1575-38L	15.62~15.87		
00-99146-1600S	C20-1600-40L	15.87~16.12	40.00	CC...0602... *NS-25060 0.9Nm NK-T7
00-99146-1625S	C20-1625-40L	16.12~16.37		
00-99146-1650S	C20-1650-41L	16.37~16.62	41.25	
00-99146-1675S	C20-1675-41L	16.62~16.87		
00-99146-1700S	C20-1700-42L	16.87~17.12	42.50	

Part No.	Type	ØD	L1	Insert Screw / Key
00-99146-1725S	C20-1725-42L	17.12~17.37	42.50	
00-99146-1750S	C20-1750-43L	17.37~17.62	43.75	
00-99146-1775S	C20-1775-43L	17.62~17.87		
00-99146-1800S	C20-1800-45L	17.87~18.12	45.00	
00-99146-1825S	C20-1825-45L	18.12~18.37		
00-99146-1850S	C20-1850-46L	18.37~18.62	46.25	
00-99146-1875S	C20-1875-46L	18.62~18.87		
00-99146-1900S	C20-1900-47L	18.87~19.12	47.50	
00-99146-1925S	C20-1925-47L	19.12~19.37		
00-99146-1950S	C20-1950-48L	19.37~19.62	48.75	
00-99146-1975S	C20-1975-48L	19.62~19.87		
00-99146-2000S	C20-2000-50L	19.87~20.12	50.00	CC...0602... *NS-25060 0.9Nm NK-T7
00-99146-2025S	C20-2025-50L	20.12~20.37		
00-99146-2050S	C20-2050-50L	20.37~20.62		
00-99146-2075S	C20-2075-50L	20.62~20.87		
00-99146-2100S	C20-2100-50L	20.87~21.12		
00-99146-2125S	C20-2125-50L	21.12~21.37		
00-99146-2150S	C20-2150-50L	21.37~21.62		
00-99146-2175S	C20-2175-50L	21.62~21.87		
00-99146-2200S	C20-2200-50L	21.87~22.12		
00-99146-2225S	C20-2225-50L	22.12~22.37		
00-99146-2250S	C20-2250-50L	22.37~22.62		
00-99146-2275S	C20-2275-50L	22.62~22.87		
00-99146-2300S	C20-2300-50L	22.87~23.12		
00-99146-2325S	C20-2325-50L	23.12~23.37		
00-99146-2350S	C20-2350-50L	23.37~23.62		
00-99146-2375S	C20-2375-50L	23.62~23.87		
00-99146-2400S	C20-2400-50L	23.87~24.12		
00-99146-2425S	C20-2425-50L	24.12~24.37		
00-99146-2450S	C20-2450-50L	24.37~24.62		
00-99146-2475S	C20-2475-50L	24.62~24.87		
00-99146-2500S	C20-2500-50L	24.87~25.12		
00-99146-2525S	C20-2525-50L	25.12~25.37		
00-99146-2550S	C20-2550-50L	25.37~25.62		

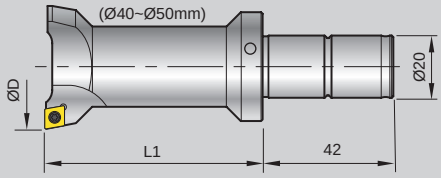
*Torque screwdriver is recommended.

99146 Quick Change High Speed EMB Boring Bar

Boring Bar Ø26~Ø39 >>



Boring Bar Ø40~Ø50 >>



- Alloy Steel Shank. Boring Depth : L1, 2~3xD
- H type with internal coolant can be ordered on request from Dia. 10mm.
  Ordering example: 00-99146-36AH.
- Other sizes are available on request.

Part No.	Type	ØD	L1	Insert Screw / Key
00-99146-26A	C20-2600-50L	25.87~26.12	50	
00-99146-27A	C20-2700-50L	26.87~27.12		
00-99146-28A	C20-2800-50L	27.87~28.12		
00-99146-29A	C20-2900-50L	28.87~29.12		
00-99146-30A	C20-3000-50L	29.87~30.12		
00-99146-31A	C20-3100-70L	30.87~31.12	70	CC...0602...
00-99146-32A	C20-3200-70L	31.87~32.12		*NS-25060 0.9Nm
00-99146-33A	C20-3300-70L	32.87~33.12		
00-99146-34A	C20-3400-70L	33.87~34.12		NK-T7
00-99146-35A	C20-3500-70L	34.87~35.12		
00-99146-36A	C20-3600-70L	35.87~36.12		
00-99146-37A	C20-3700-70L	36.87~37.12		
00-99146-38A	C20-3800-70L	37.87~38.12		
00-99146-39A	C20-3900-70L	38.87~39.12		

Part No.	Type	ØD	L1	Insert Screw / Key
00-99146-40A	C20-4000-70L	39.87-40.12	70	
00-99146-41A	C20-4100-70L	40.87-41.12		
00-99146-42A	C20-4200-70L	41.87-42.12		
00-99146-43A	C20-4300-70L	42.87-43.12		
00-99146-44A	C20-4400-70L	43.87-44.12		CC...0602...
00-99146-45A	C20-4500-70L	44.87-45.12		*NS-25060 0.9Nm
00-99146-46A	C20-4600-70L	45.87-46.12		NK-T7
00-99146-47A	C20-4700-70L	46.87-47.12		
00-99146-48A	C20-4800-70L	47.87-48.12		
00-99146-49A	C20-4900-70L	48.87-49.12		
00-99146-50A	C20-5000-70L	49.87-50.12		

*Torque screwdriver is recommended.

High Speed Boring Bar Kit >>

Part No.	Contents	
00-99146-SB32H-05SET	SB32-146-31 Weldon Shank	Boring head shank: 1pc Boring bar: any 5 pcs from Ø5~Ø50 Key: 3~5 pcs Plastic box: 1pc
00-99146-BT30-05SET	BT30H Boring head shank	
00-99146-BT40-05SET	BT40H Boring head shank	
00-99146-BT50-05SET	BT50H Boring head shank	
00-99146-CAT40-05SET	CAT40H Boring head shank	
00-99146-HSK63A-05SET	HSK63A Boring head shank	



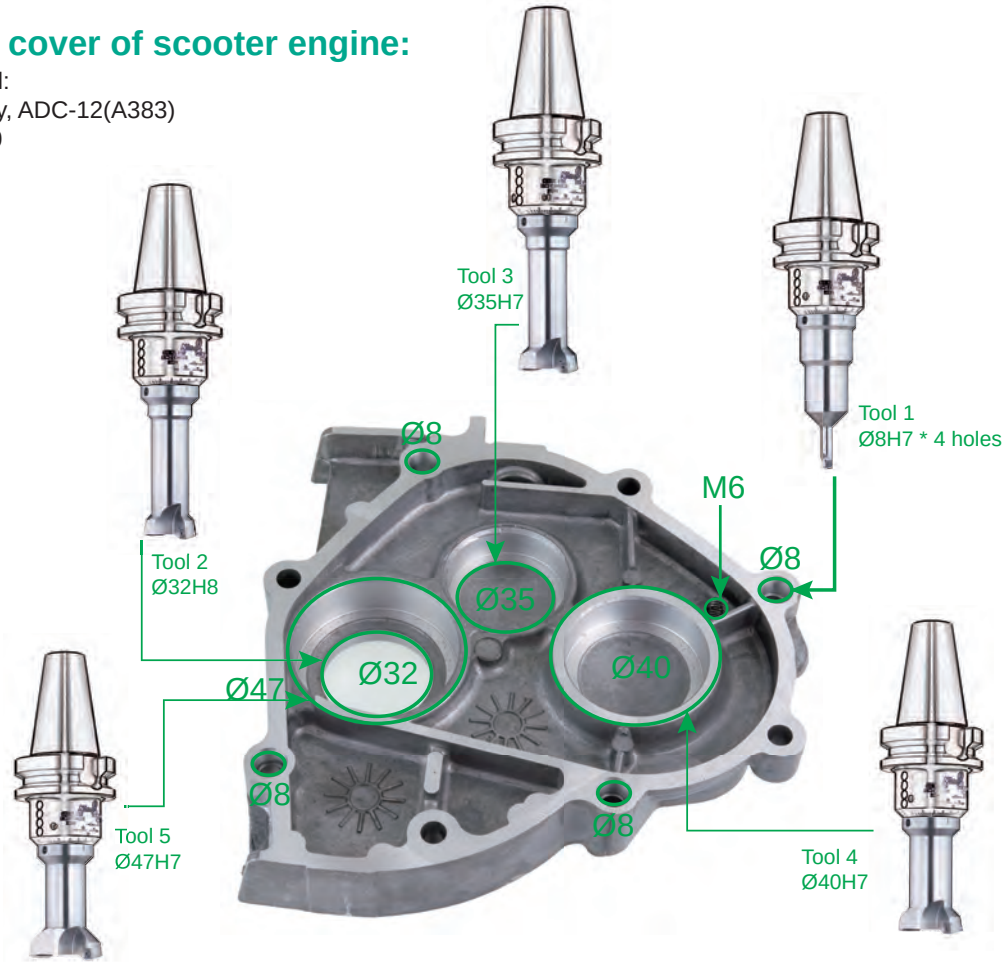
(Insert is not included, please order seperately)

- Note: BT50 boring head shank is packed in a separate box.

Application Example

Machining a cover of scooter engine:

Workpiece material:
Die casting, Al-alloy, ADC-12(A383)
Spindle Size: BT40



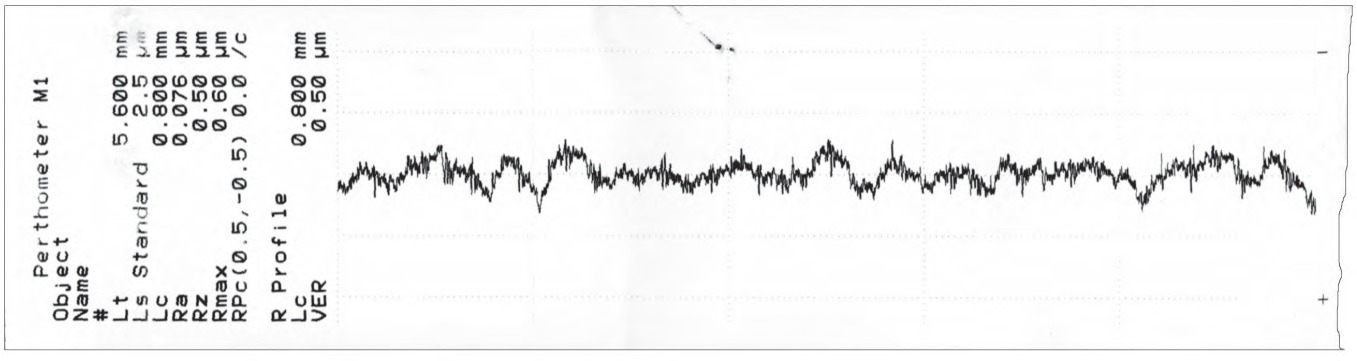
TOOL LIST by Nine9 Boring Bar 99146-series :

No.	Boring Bar	Grade of insert	Dia. mm	Depth	r.p.m.	F = mm/min.	Machining time
1	00-99146-08A	CCGT040102 NC30	Ø8H7	8 mm	8000	400	1.2 sec.
2	00-99146-32A	CCFT060204HP NC9031	Ø32H8	8 mm	2985	209	2.3 sec.
3	00-99146-35A		Ø35H7	12 mm	2730	191	3.8 sec.
4	00-99146-40A		Ø40H7	15 mm	2400	168	5.4 sec.
5	00-99146-47A		Ø47H7	15 mm	2030	142	6.4 sec.

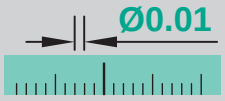
Example >>



Material	Vc m/min.	f mm/rev.	Roughness			Tool holder	Insert
			Ra	Rz	Rmax		
Al alloy, 6061	150	0.2	0.076μm	0.50μm	0.6μm	99146-BT40-26A	CCGH0602U NC9036

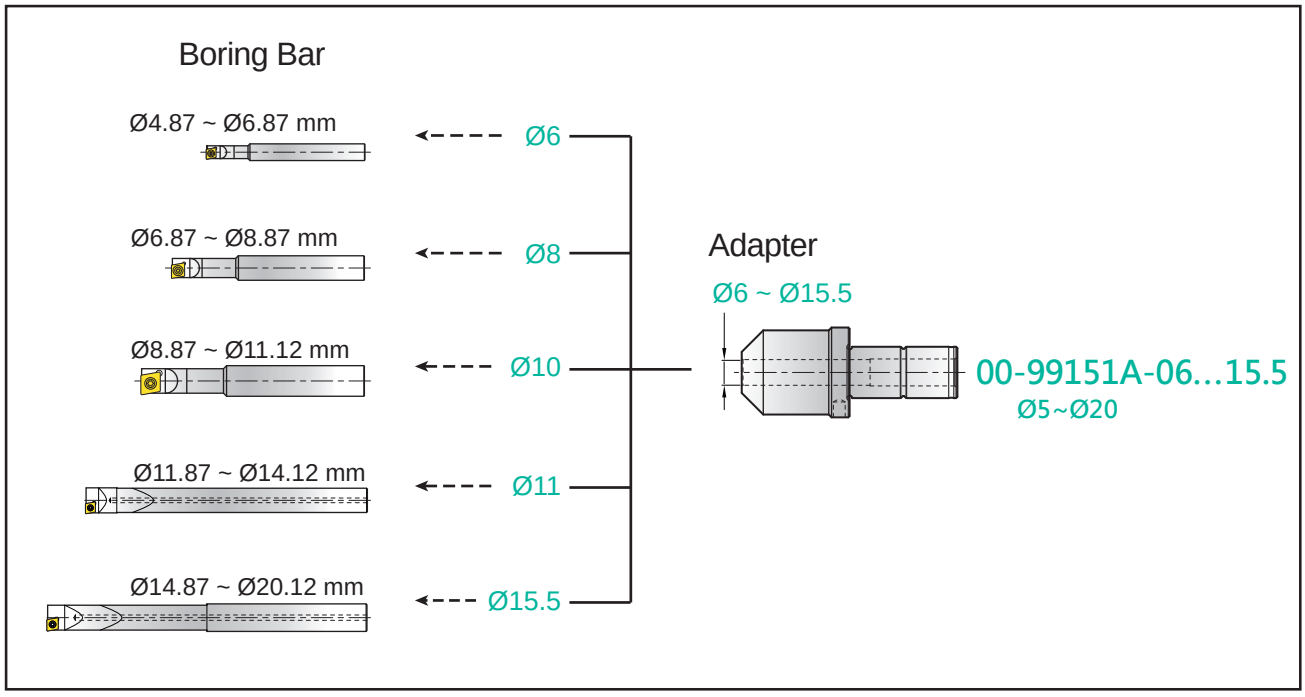


99151 Deep hole boring 4~6XD





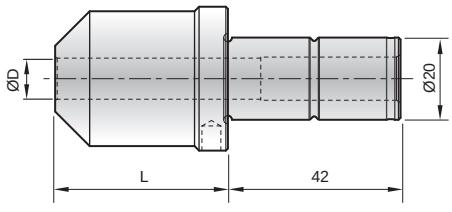
99151 System >>



Adapter >>

- Economical solution of small dia. boring bar.



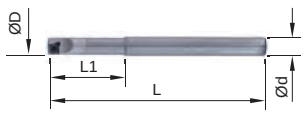
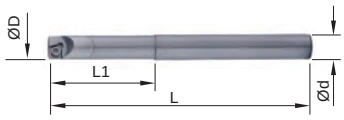
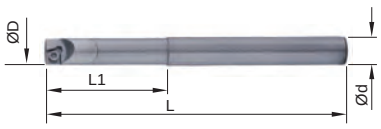
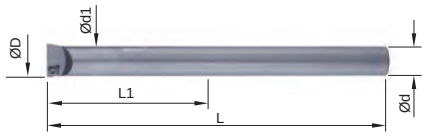
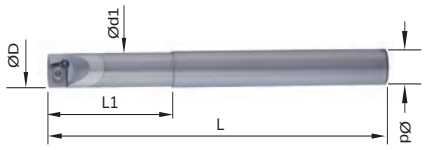
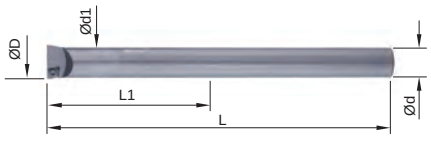


Part No.	Type	ØD	L
00-99151A-06	C20-ID06	6	52
00-99151A-08	C20-ID08	8	49
00-99151A-10	C20-ID10	10	42
00-99151A-11	C20-ID11	11	21.5
00-99151A-15.5	C20-ID15.5	15.5	21.5

99151 Deep hole boring 4~6XD

Boring Bar Ø5~Ø20 >>

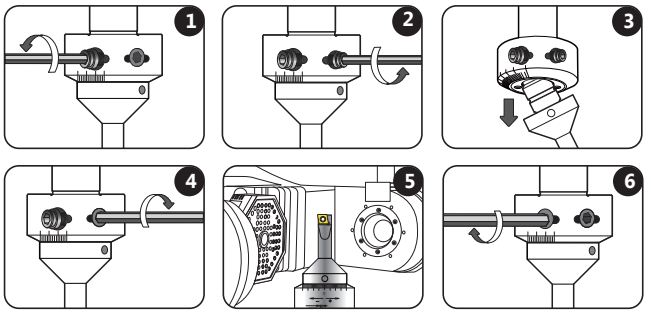
- Solid Carbide Shank
- Boring Depth : L1, 4~6xD

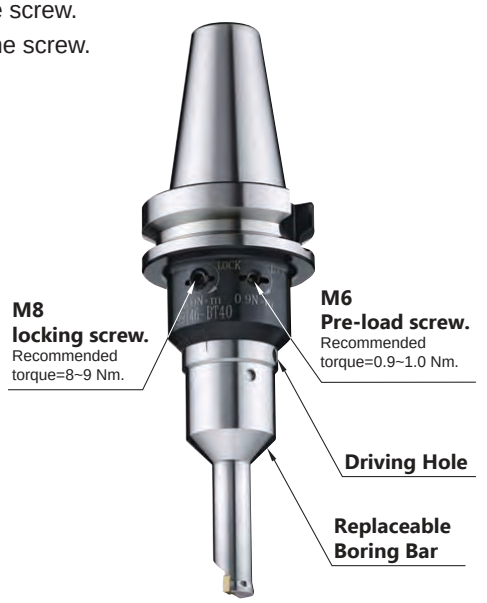
Part No.	Type	ØD	Ød	Ød1	L1	L	Insert Screw / Key	Fig.
00-99151-0500W	C06-0500-20L	4.87~5.12	6	-	20	70	CCGT030102 *NS-16030 / 0.4Nm NK-T6	
00-99151-0525W	C06-0525-20L	5.12~5.37		-				
00-99151-0550W	C06-0550-22L	5.37~5.62		-	22			
00-99151-0575W	C06-0575-22L	5.62~5.87		-				
00-99151-0600W	C06-0600-24L	5.87~6.12		-	24			
00-99151-0625W	C06-0625-24L	6.12~6.37		-				
00-99151-0650W	C06-0650-26L	6.37~6.62		-	26			
00-99151-0675W	C06-0675-26L	6.62~6.87		-				
00-99151-0700W	C08-0700-28L	6.87~7.12	8	-	28	85	CCGT040102 *NS-20036 / 0.6Nm NK-T6	
00-99151-0725W	C08-0725-28L	7.12~7.37		-				
00-99151-0750W	C08-0750-30L	7.37~7.62		-	30			
00-99151-0775W	C08-0775-30L	7.62~7.87		-				
00-99151-0800W	C08-0800-32L	7.87~8.12		-	32			
00-99151-0825W	C08-0825-32L	8.12~8.37		-				
00-99151-0850W	C08-0850-34L	8.37~8.62		-	34			
00-99151-0875W	C08-0875-34L	8.62~8.87		-				
00-99151-0900W	C10-0900-36L	8.87~9.12	10	-	36	110	CC...0602... *NS-25045 / 0.9Nm NK-T7	
00-99151-0925W	C10-0925-36L	9.12~9.37		-				
00-99151-0950W	C10-0950-38L	9.37~9.62		-	38			
00-99151-0975W	C10-0975-38L	9.62~9.87		-				
00-99151-1000W	C10-1000-40L	9.87~10.12		-	40			
00-99151-1025W	C10-1025-40L	10.12~10.37		-				
00-99151-1050W	C10-1050-42L	10.37~10.62		-	42			
00-99151-1075W	C10-1075-42L	10.62~10.87		-				
00-99151-1100W	C10-1100-44L	10.87~11.12		-	44			
00-99151-1200WS	C11-1200-120L	11.87~12.12	11	11	70	120	CC...0602... *NS-25045 / 0.9Nm NK-T7	
00-99151-1300WS	C11-1300-120L	12.87~13.12	11	-				
00-99151-1400WS	C11-1400-120L	13.87~14.12		-				
00-99151-1500W	C15.5-1500-180L	14.87~15.12	15.5	14	90	180		
00-99151-1600W	C15.5-1600-180L	15.87~16.12	15.5	15	90	180	CC...0602... *NS-25060 / 0.9Nm NK-T7	
00-99151-1700W	C15.5-1700-180L	16.87~17.12		-	100			
00-99151-1800W	C15.5-1800-180L	17.87~18.12		-				
00-99151-1900W	C15.5-1900-180L	18.87~19.12		-				
00-99151-2000W	C15.5-2000-180L	19.87~20.12		-				

*Torque screwdriver is recommended.

Procedures For Assembly

1. Use 4 mm allen-key to loosen locking screw M8, take care not to remove the screw.
2. Use 3 mm allen-key to loosen pre-load screw M6, take care not to remove the screw.
3. Remove the original boring bar and insert the new boring bar.
4. Tighten the M6 pre-load screw. Recomended torque = 0.9 ~ 1.0Nm.
5. Measure the boring diameter of the boring bar using tool presetter and adjust it to the required diameter.
6. Tighten the M8 locking screw. Recomended torque = 8 ~ 9Nm.





Procedures For Adjustment

On Tool Presetter

1. Loosen M8 locking screw.
2. Set the boring bar at the neutral position. (Step 1)
3. Measure the boring diameter using the tool presetter and compare with the required diameter. (Step 2)
4. If boring diameter is too big or too small, please put an allen-key into the adjusting driving hole.
   Turn to " + " to increase and turn to " - " to reduce boring diameter. (Step 3 and 4)
5. Tighten M8 locking screw.

(Step 1)



(Step 2)



(Step 3)



(Step 4)



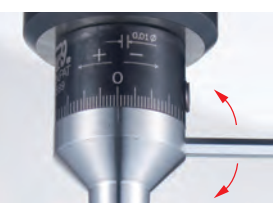
To Increase Diameter

To Reduce Diameter

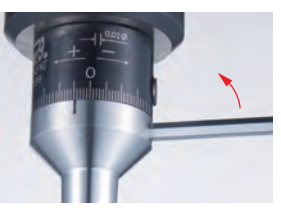
On Milling Machine and Machining Centers

1. Set the boring bar at the neutral position. (Step 1)
2. Tighten M8 locking screw.
3. Test cut on work piece, about 3-5mm deepth on the machine.
4. Measuring boring diameter of workpiece and compare with required diameter.
5. If boring diameter is too big or too small, loosen M8 locking screw, please put an allen-key into the adjusting driving hole.
   Turn to " + " to increase and
   turn to " - " to reduce boring diameter. (Step 2 and 3)
6. Tighten M8 locking screw. (Step 4)

(Step 1)

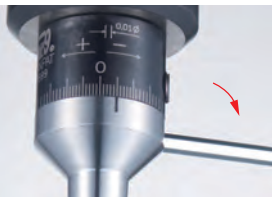


(Step 2)



To Increase Diameter

(Step 3)



(Step 4)



To Reduce Diameter

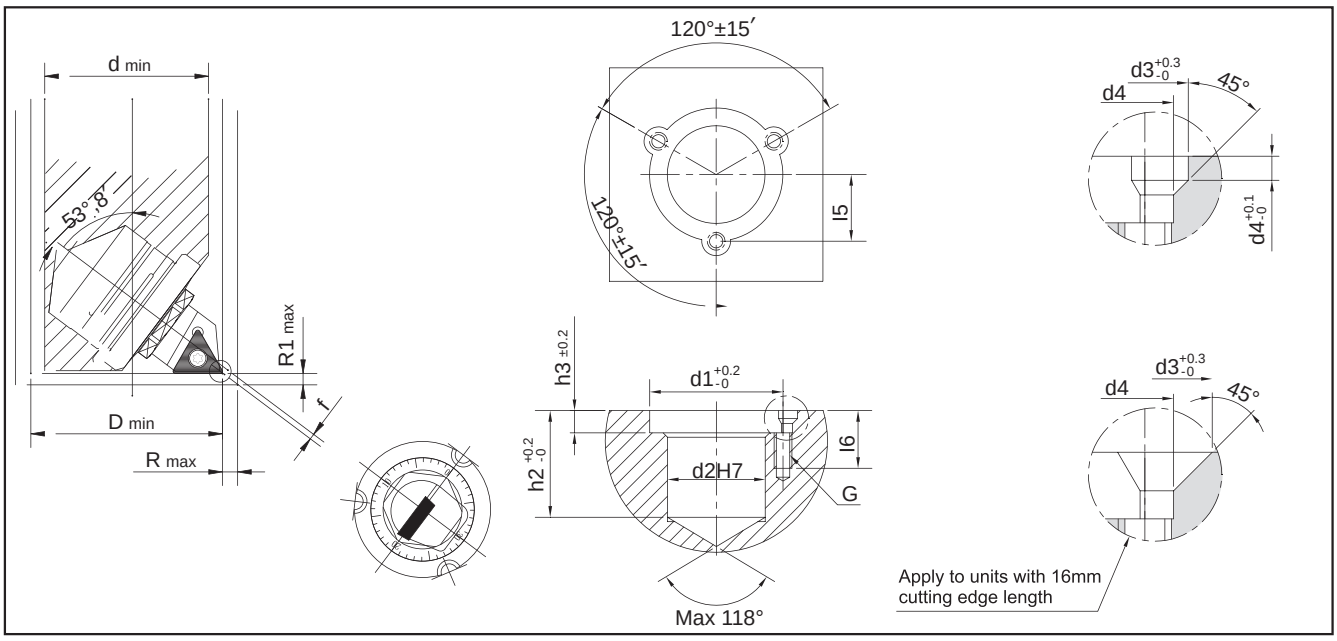
Micro Adjustable Fine Boring Unit

Angular mounting 90°

- Insert to be used:
  CCMT, CCMW, CCGH, CCGT, CCGX,
  TCMT, TCMW, TCGT, TCGX
- Maximum range can be checked by end of spanner.







Parts No.			Insert Radius	Dimensions / mm					Mounting Dimensions / mm									
				dmin	Dmin	Rmin	R1max	f	h2	h3	h4	d1	d2	d3	d4	G	l6	l5
3106		06	0.2	22	25.7	2	1.5	0.40	11.5	2.8	1.6	19	16	4.6	3.2	M3	10	9.65 ±0.02
			0.4		25.4			0.40										
			0.8		24.8			0.45										
3209	09		0.2	28.5	33.4	2.8	2.1	0.95	15.5	4	1.6	25	20	4.6	3.2	M3	10	12.5 ±0.02
			0.4		33.1			1.0										
			0.8		32.5			1.1										
3311	11		0.2	38	42.9	4.8	3.6	1.15	24	5	1.8	30	22	6.5	4.3	M4	15	15.4 ±0.02
			0.4		42.6			1.2										
			0.8		42.0			1.3										
3416	16		0.2	55	60.6	8	6	1.3	33	6.3	-	46	32	11.9	5.4	M5	20	23 ±0.02
			0.4		60.0			1.4										
			1.2		59.4			1.5										

Spare parts for setting and clamping

Item			Spanner	Mounting Screw	Insert Screw	Mounting Screw Key	Insert Screw Key	Mounting Screw Key
3106		06	3106-SP	NS-30072	NS-25060	NK-T9	NK-T7	-
3209	09		3209-SP	NS-30072	191.958	NK-T9	NK-T7	-
3311	11		3311-SP	CS4011	NS-25060	NK-T15	NK-T7	-
3416	16		3416-SP	3416H	NS-35080	-	NK-T15	3146-J

Boring Tool Precisely Ground Inserts

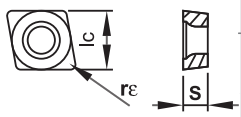
NC30 : • Universal grade for casting iron, carbon steel, alloy steel, stainless steel.

NC2032 : • For high speed cutting of casting iron.

NC2033 : • Good for carbon steel, alloy steel, stainless steel.

NC9036 : • long tool life.
• Good for Al, Al-alloy, Copper and non-ferrous metal.

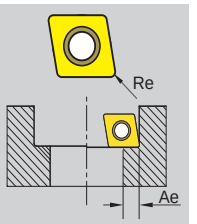
XP9001 : • Super finishing insert with special specified cutting width 0.15mm radius for high feed rate.
• Good for Al, Al-alloy and non-ferrous metal.

Parts No.		Coating	Grade		Dimensions			Screw	Key
					Ic	S	Re		
CCGT030102	NC30	TiAlN	K20F		3.5	1.4	0.2	*NS-16030 0.4Nm	NK-T6
	NC9036	DLC							
CCGT040102	NC30	TiAlN	K20F		4.3	1.8	0.2	*NS-20036 0.6Nm	NK-T6
	NC9036	DLC							
CCGH0602U	XP9001	Uncoated	K20F				-		
CCFT060204	NC2033	TiAlN	K20F		6.35	2.38	0.4	*NS-25045 0.9Nm	NK-T7
	NC9036	DLC							
CCFW060204	NC2032	AlTiN	K20F				0.4		

*Torque screwdriver is recommended.

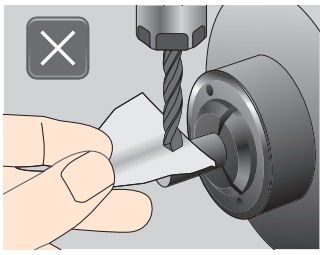
Cutting Data

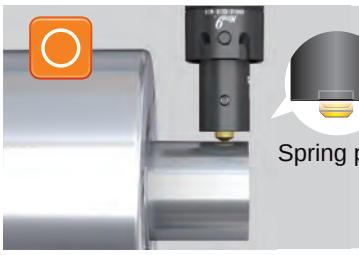
Spindle speed $S = \frac{Vc \times 1000}{\pi \times D}$ r.p.m. Feed rate: f x S mm/min.



Workpiece Meterial		Cutting conditions or surface finishes	Cutting Speed Vc(m/min.)	feed rate f (mm/rev.)	Re0.2	Re0.4	Grade of Insert
					Ae (mm)		
P	Carbon Steel	Regular cutting	120-150-200	0.05-0.07-0.10	0.05	0.1	NC2033
		Interrupted cutting	100-120-140	0.04-0.05-0.08	0.05	0.1	NC30
	Alloy Steel	Regular cutting	100-120-140	0.05-0.07-0.10	0.05	0.1	NC2033
		Interrupted cutting	80-100-120	0.04-0.05-0.08	0.05	0.1	NC30
M	Stainless steel	Regular cutting	80-100-120	0.05-0.07-0.10	0.05	0.1	NC2033
		Interrupted cutting	70-80-100	0.05-0.07-0.10	0.05	0.1	NC30
K	Cast Iron	Regular cutting	80-100-120	0.05-0.07-0.10	0.05	0.1	NC2032 / NC30
N	Non-ferrous metal	Regular cutting	150-200-300	0.05-0.07-0.10	0.05	0.1	NC9036
		Super mirror finish	150-200-300	0.15-0.20-0.25	0.15		XP9001
H	Hardened Steel <HRC 50	Regular cutting	80-100-120	0.04-0.06-0.08	0.05	0.1	NC30

Accessories





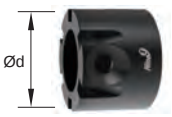
Quick and simple tool length setting!

Ergo tool length setter TP >>

- Ergo setter is an easy tool length recorder while setting the tool length on swiss type automatic lathe and CNC turning centers.
- Reduce machine downtime, prevent insert and workpiece from damage.

Code	Parts No.	Illustration	L-Key
16-TP0001	00-99816-TP	 Focusing edge (For Tool Presetter)	NK-LW15 (2 Nm)

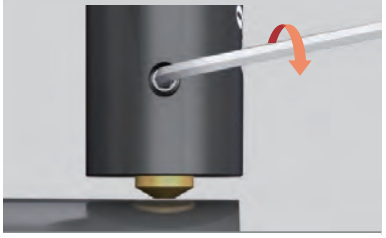
Spare >>

Set			Ergo Nut				High Strength Ergo Pin			L-Key	Ergo Spanner
 * Nut, pin & L-key are included.			 Ød				 L				
ER	Parts No.		Parts No.	Ød	Pitch	Torque	Parts No.	L	Torque	Parts No.	Parts No.
ER16	00-99816-M19S		00-99816-M19	25	M19xP1.0	30 Nm	NS-50025	25	5 Nm	NK-LW3	00-99816-SP28
	00-99816-M22S		00-99816-M22	28	M22xP1.5	30 Nm	NS-50028	28	5 Nm		

Setting process >>

• Step-1

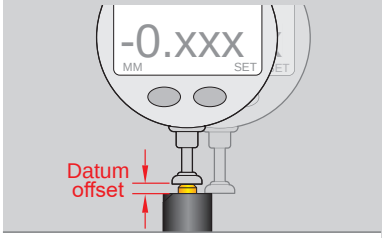
Set Temporary Position



1-1: Move the setter tip to touch the center-top of workpiece.
1-2: Press spring pin 1~2 mm down.
1-3: Tighten screw to fix spring pin, and get a temporary length of setter.
1-4: Input the temporary length value to the CNC controller.

• Step-2

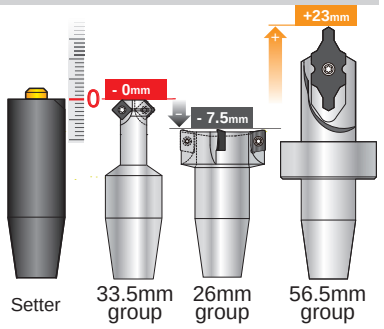
Get The Datum Position



2-1: The offline measures the datum offset of setter by height gauge.
2-2: Input datum offset to CNC controller.

• Step-3

Input Tool Length Offset



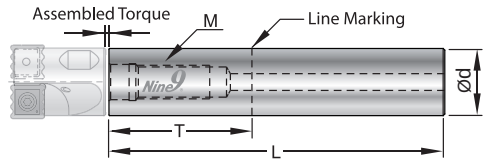
3-1: Choose an Ergo tool to install, and input the offset value to CNC controller directly.

Extension Bar

For ACE Spot Drill, NC Spot Drill, Chamfer Mill, NC Helix Drill, Power Mill.

Steel Type >>

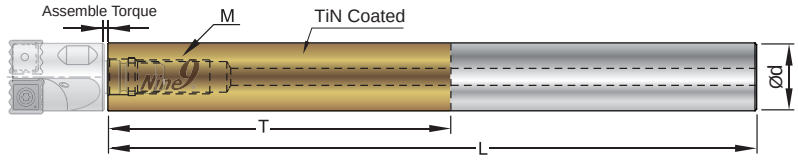
- T is the maximum overhang length.
- With internal coolant hole.



Part No.	Type	Ød	T	L	M	Assemble Torque
00-99801-10S	BC10-075M05S	10	25	75	M5xP0.8	6.5 Nm
00-99801-12S	BC12-075M06S	12	25	75	M6xP1.0	11.0 Nm
00-99801-14S	BC14-090M08S	14	30	90	M8xP1.25	25.0 Nm
00-99801-16S	BC16-090M08S	16	35	90	M8xP1.25	25.0 Nm
00-99801-18S	BC18-100M10S	18	40	100	M10xP1.5	50.0 Nm
00-99801-20S	BC20-100M10S	20	40	100	M10xP1.5	50.0 Nm
00-99801-25S	BC25-120M12S	25	50	120	M12xP1.75	60.0 Nm

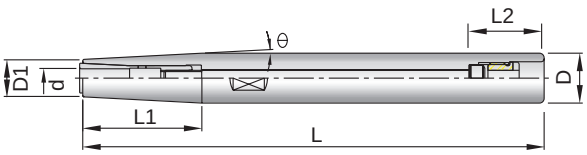
Solid Carbide Type >>

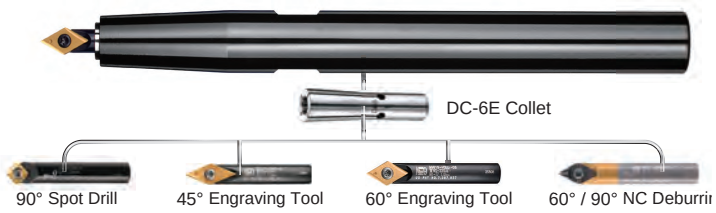
- T is the maximum overhang length.
- With internal coolant hole.



Part No.	Type	Ød	T	L	M	Assemble Torque
00-99801-08W	BC08-075M04W	8	25	75	M4xP0.7	3.5 Nm
00-99801-10W	BC10-100M05W	10	50	100	M5xP0.8	6.5 Nm
00-99801-12W	BC12-100M06W	12	60	100	M6xP1.0	11.0 Nm
00-99801-14W	BC14-120M08W	14	70	120	M8xP1.25	25.0 Nm
00-99801-16W	BC16-150M08W	16	80	150	M8xP1.25	25.0 Nm
00-99801-18W	BC18-150M10W	18	90	150	M10xP1.5	50.0 Nm
00-99801-20W	BC20-200M10W	20	100	200	M10xP1.5	50.0 Nm
00-99801-25W	BC25-200M12W	25	125	200	M12xP1.75	60.0 Nm

DC Slim Chuck





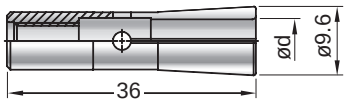
Parts No.	Type of Holder	d	L	L1	L2	D	D1	θ	Collet	Back Screw	Stop Screw	Hexagon Key	Stop Nut
0-329090-212	ST12-DC6-120	2~6	120	40	-	12	14	--	DC6	M5 * L95	--	0-301940~642	TP-M12
-222	ST16-DC6-150	2~6	150	38	24	16	14	3°		M5 * L100	OP-M10		--
-232	ST20-DC6-200	2~6	200	70	74	20	14	3°		M5 * L100	OP-M10		--
-242	ST25-DC6-250	2~6	250	115	124	25	14	3°		M5 * L100	OP-M10	0-301940~643	--

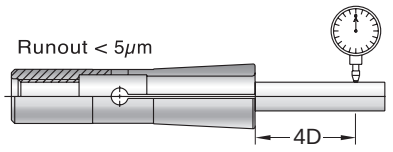
* Stop nut is applied when clamping and unclamping tools.

DC-6E Collet >>

- The design of DC-E collets is emphasized on increasing the clamping force of end mills.

Parts No.	Size(mm)
0-300090-203	3.0
0-300090-204	4.0
0-300090-206	6.0





Notes

No Need To Choose
Nine9 Does It All

Always Better
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